
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) models have become a central mechanism for scaling large language models, yet the representational mechanisms by which they diverge from dense networks remain insufficiently characterized. Prior work has shown that dense models exploit *superposition*, representing more features than ambient dimensions permit orthogonally, and that this behavior varies systematically with feature sparsity and feature importance. We show that MoE models are not mechanistically well described by the same explanatory axis: neither feature sparsity nor feature importance induces the discontinuous phase transitions observed in dense settings, whereas network sparsity, understood as the ratio of active to total experts, more faithfully organizes MoE behavior. To make this distinction measurable, we introduce metrics for quantifying superposition across experts and find that greater network sparsity is associated with greater *monosemanticity*. We further propose a definition of expert specialization grounded in monosemantic feature representation rather than aggregate load balancing, demonstrating that, under appropriate initialization, experts naturally organize around coherent feature combinations. Taken together, these results suggest that network sparsity in MoEs can support more interpretable representations without commensurate performance degradation, thereby complicating the common presumption that interpretability and capability are necessarily antagonistic.

1 INTRODUCTION

Mixture of Experts (MoEs) have become increasingly prevalent in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), largely because they deliver favorable computational scaling and empirical performance gains (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work has refined routing mechanisms, including Expert-Choice routing, and improved training stability and transfer, as exemplified by ST-MoE (Zhou et al., 2022; Zoph et al., 2022). Complementary theoretical and empirical analyses further indicate that learnable routers can recover latent cluster structure in the data, thereby offering a partial explanation for the emergence of expert specialization (Dikkala et al., 2023). Nevertheless, despite their extensive deployment, MoEs remain comparatively opaque from the standpoint of mechanistic interpretability.

Interpretability-oriented approaches have correspondingly attempted to render expert behavior more transparent. Yang et al. (2025b) proposes MoE-X, which promotes sparsity-aware routing and employs wide, ReLU-based experts to reduce polysemanticity, while Park et al. (2025) introduce MONET, scaling expert counts in order to enable capability editing through expert activation. Although these contributions illuminate important architectural interventions, they leave unresolved the more mechanistic questions of how MoEs encode features, how the presence of experts changes superposition, and whether specialization can arise without additional regularization. Consistent with this gap, Mu & Lin (2025) survey MoE research and identify mechanistic interpretability as a central open challenge.

A central obstacle for neural network interpretation is superposition, the representational regime in which a model encodes more features than it has available dimensions. This

regime permits networks to compress many sparsely active features into comparatively few neurons, but the resulting neurons become polysemantic and consequently more difficult to interpret mechanistically.

MoE architectures introduce an additional axis into this problem: network sparsity. Whereas dense models activate all neurons for every input, MoEs activate only a fraction of their total parameters (Shazeer et al., 2017). Consequently, although dense models exploit feature sparsity by packing numerous sparse features into shared representational directions, MoEs can allocate computation more selectively and may, in favorable cases, dedicate entire experts to particular feature combinations.

We therefore investigate three linked questions: (1) whether MoEs exhibit less superposition than parameter-comparable dense counterparts, (2) whether the superposition of a particular feature within MoE experts undergoes the same discrete phase changes over relative importance and global feature sparsity that appear in dense models, and (3) whether expert specialization is more productively understood through feature representation than through load balancing alone.

We address these questions with simple models that extend the framework of Elhage et al. (2022) to the MoE setting. Our principal contributions are threefold: (1) unlike dense models, MoEs do not display sharp phase changes, but instead exhibit increasingly continuous representational transitions as network sparsity grows; (2) MoEs consistently display greater monosemanticity, equivalently less superposition, than dense models with matched active and total parameter counts, with individual experts encoding features more cleanly; (3) we formulate an interpretability-oriented definition of expert specialization based on monosemantic feature representation, showing that experts can organize around coherent feature combinations rather than merely around arbitrary load-balancing patterns.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis posits that networks encode concepts as directions in activation space (Park et al., 2024); however, the inventory of interpretable features frequently exceeds the dimensionality available for mutually orthogonal representation. Elhage et al. (2022) formalized this mismatch as superposition, showing that dense models use non-orthogonal feature packing to increase representational capacity, albeit at the cost of polysemanticity. Although superposition is theoretically efficient (Scherlis et al., 2025), it often necessitates post-hoc disentanglement methods, including Sparse Autoencoders, to recover monosemantic features (Bricken et al., 2023). Recent work has further examined how data correlations shape superposition (Prieto et al., 2025) and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures and show that network sparsity, rather than feature sparsity alone, substantially governs their representational strategies.

Interpretability of MoEs. Although MoEs have become a standard strategy for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations remains less developed than for dense architectures. Existing analyses have primarily emphasized macroscopic phenomena, including routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023). More recent interpretability-oriented approaches attempt to induce specialization through architectural constraints, such as pre-defined concept routing (Yang et al., 2025b), or by scaling expert counts to match vocabulary sizes (Park et al., 2025). Yet these approaches often prioritize capability editing over an explanation of intrinsic feature geometry. We address this gap by analyzing how experts modulate superposition, by showing that MoEs exhibit greater monosemanticity than dense models, and by proposing a feature-based account of expert specialization.

3 BACKGROUND

A primary objective of mechanistic interpretability is the reverse engineering of neural networks, and one influential approach decomposes model representations into human-interpretable concepts conventionally called ‘features’. These features are frequently assumed to be linear, in the sense that any hidden state h can be expressed as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ denotes the direction associated with feature i , α_i denotes that feature’s activation strength, roughly corresponding to the extent to which feature i is present in the input, and F denotes the set of all represented features.

The *superposition hypothesis* holds that models can represent many more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Achieving such overcomplete representation requires the feature vectors in F to be packed non-orthogonally within the latent space. When two features exhibit interference, $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say that they are in *superposition*. This arrangement can be tolerable when features are sparsely active, as is characteristic of much online text data, but it imposes an interpretability cost because α_i may contain spurious activation attributable to features other than feature i itself.

Monosemanticity is usually defined at the level of individual neurons, where a neuron’s activation corresponds cleanly to a single α_i , or equivalently where features are basis-aligned. If features are orthogonal, and therefore not in superposition, but are not basis-aligned, individual neurons may remain *polysemantic* even though feature interference is minimal. In this paper, our emphasis is therefore on reducing superposition rather than enforcing basis alignment; for brevity, when we describe models, experts, or features as “more monosemantic,” we mean that they exhibit less superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are often informally conceptualized as collections of dense models, with each expert treated as an independent dense network. Whether experts actually acquire feature representations that resemble those of dense models, however, is not obvious. We examine this issue by directly comparing the superposition patterns learned by MoE and dense architectures.

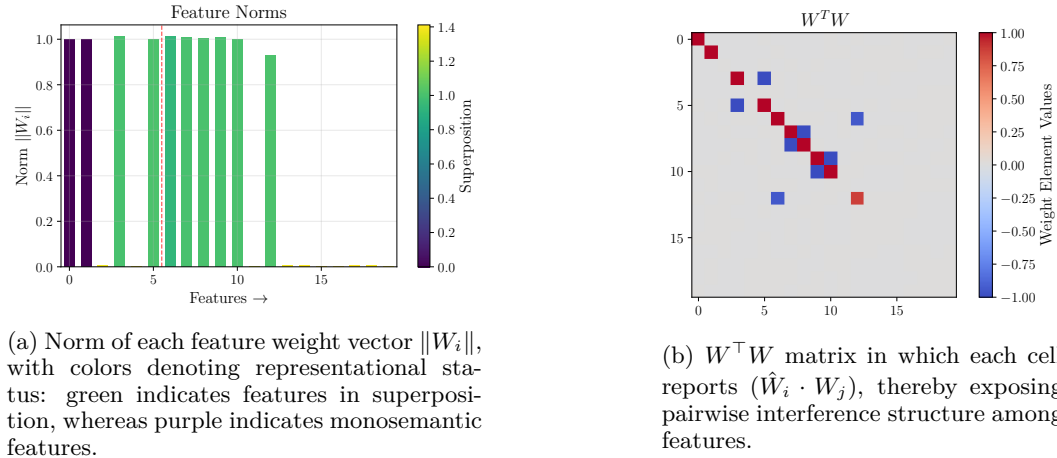


Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, using importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. Color encodes superposition as $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

Our objective is to examine how a MoE projects a high-dimensional vector, $x \in \mathbb{R}^n$, into lower-dimensional expert representations, $h \in \mathbb{R}^m$, while still permitting accurate reconstruction. This experimental design extends the framework of Elhage et al. (2022) from dense autoencoders to the MoE setting.

The input distribution. The input vector x represents activations from an idealized disentangled model in which each coordinate x_i corresponds to a distinct and independent feature, effectively constituting a perfectly neuron-aligned and monosemantic representation. We draw $x_i \sim U(0, 1)$ except under a prescribed sparsity level $S \in [0, 1]$, for which $P(x_i = 0) = S$. Because not all features are assumed to contribute equally to the objective, each feature x_i is assigned a scalar importance I_i that weights the reconstruction loss. To isolate phase transitions associated with a single feature, we additionally scale the magnitude of the final feature by $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. Together, I_i and r modulate feature signal strength and thereby allow us to test expert sensitivity to feature magnitude.

Model Architecture. The MoE contains E experts, with each expert e parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, where $W^r \in \mathbb{R}^{E \times n}$, assigns each input to the top- k experts. Each active expert maps the input into a lower-dimensional hidden state $h^e = W^e x$ and then produces a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The model output is the weighted combination of active expert reconstructions, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where w_e denotes the renormalized gating weights.

We train all models using an L2 reconstruction loss weighted by feature importances, I_i , namely $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, where expert collapse would otherwise confound the phase analysis by allowing experts and routers to settle into minima in which some experts are ignored, we add the standard auxiliary load-balancing loss (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the number of experts, f_e is the fraction of batch samples routed to expert e , P_e is the mean gating probability assigned to expert e across the batch, and $\alpha = 0.01$ sets the penalty strength. We deliberately omit auxiliary load balancing in Section 4 so that the measured superposition patterns reflect the MoE architecture’s intrinsic bias rather than an externally imposed balancing objective.

4.2 MEASURING FEATURE CAPACITY

To compare feature representations across architectures, we examine two basic quantities: *representation strength* and *interference* with other features. Representation strength is measured by the norm of the feature weight vector in expert e , $\|W_i^e\|$, which indicates the extent to which feature i is encoded by that expert: $\|W_i^e\| \approx 1$ corresponds to full representation, whereas a norm near zero indicates that the feature has not been learned. We visualize the interference between feature i and other features within expert e using the Gram matrix $W^\top W$, whose off-diagonal elements encode pairwise interference.

As Figures 1a and 2a show, the dense model and the MoE represent a comparable number of features, 10 versus 8, with similar norms under an equal total-parameter budget ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$). Although individual MoE experts can still exhibit local superposition, as in Expert 0 of Fig 2b, the global interference structure is sharply partitioned. Unlike the dense model, shown in Figure 1b, where any feature can interfere with any other, the MoE induces a block-diagonal organization in which features routed to distinct experts have zero mutual interference. This pattern indicates that MoEs allocate representational capacity by partitioning feature space and thereby constraining the global extent of interference.

Expert Feature Dimensionality. To characterize how MoEs allocate limited representational capacity differently from dense models, we measure feature dimensionality, the

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0, 1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^\top W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

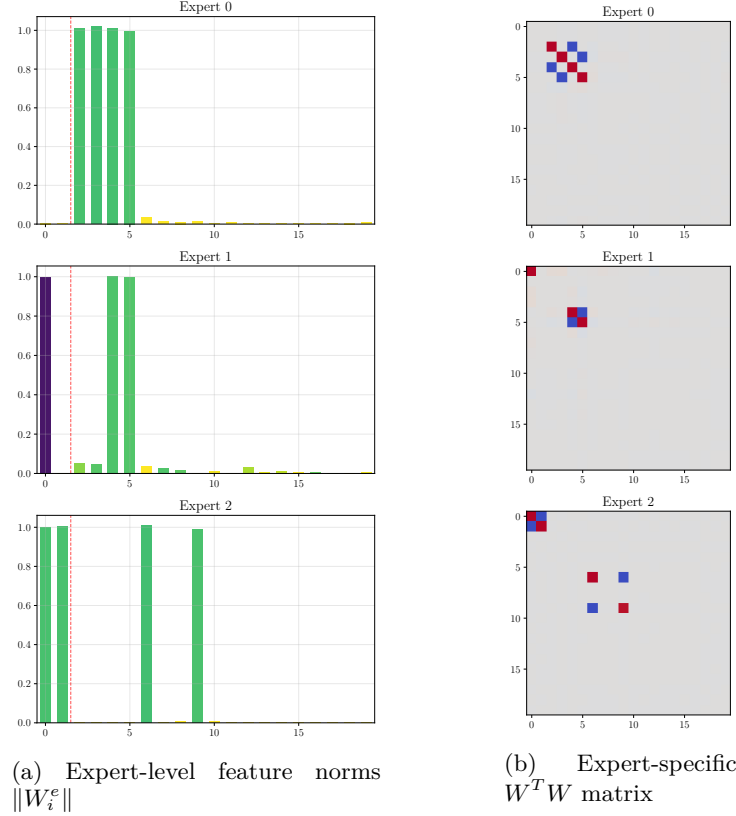


Figure 2: Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under top- $k = 1$ routing, using importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

“fraction of a dimension” effectively assigned to a particular feature (Elhage et al., 2022). For feature i , its dimensionality in expert e is defined as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e is bounded between zero, corresponding to an unlearned feature, and one, corresponding to monosemantic representation. The aggregate MoE capacity can therefore be written as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. When features are “efficiently packed” into a model’s representational space, the summed dimensionality of the represented features approximates the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE, the analogous relation is $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, indicating that, for a fixed total-parameter budget, MoEs pack features with comparable efficiency to dense models.

Features per Dimension & Network Sparsity. Because both dense and MoE models efficiently pack features into their representational spaces, we compare architectures by measuring the number of features represented per dimension. This quantity provides a direct measure of superposition and allows us to assess how increasing the number of MoE experts changes superposition across feature sparsity regimes. Whenever features per dimension exceeds one, the model is necessarily representing more features than dimensions and is therefore operating in superposition. For a MoE, we define features per dimension as

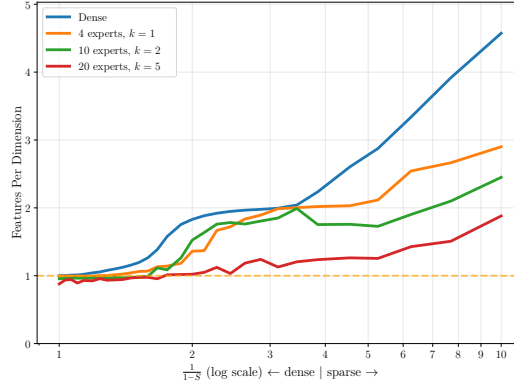


Figure 3: Features per dimension as a function of inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the greatest superposition, which declines as expert count increases: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameter counts and similar k/E . The dashed line at 1.0 denotes monosemantic representation.

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ denotes the Frobenius norm and p_e denotes the expected usage probability of expert e over a batch of input samples, equivalently the average renormalized gating weight after top- k routing.

For MoEs with the *same* total number of parameters as the dense model, the dense model exhibits more features per dimension, and hence greater superposition, as shown in Figure 3. This indicates that the dense model uses superposition to encode a larger total number of learned features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE tends to saturate near the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Moreover, as the number of MoE experts increases while total parameters and the ratio k/E remain approximately fixed, features per dimension decreases, indicating reduced superposition. *The greater the number of experts, the less superposition in the model.* Equivalently, feature representations become increasingly monosemantic as expert count rises. The dense model’s greater use of superposition also permits consistently lower reconstruction loss than the MoEs, as shown in Figure 6 in Appendix A.1, with a loss difference of $\sim 0.03 - 0.08$ at any given sparsity level. However, as expert count increases, MoE losses become consistently comparable to dense-model losses. Appendix A.2 provides theoretical intuition for this pattern.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute those features across experts in a manner that substantially reduces interference. This observation suggests that network sparsity primarily reshapes the allocation of learned features rather than simply changing their number. We therefore examine how input-distribution properties, including feature sparsity and feature importance, govern this allocation and whether they induce physics-inspired *phase changes* in representation.

Because models possess finite representational capacity, each feature must be ignored, represented in superposition, or encoded monosemantically. A phase change refers to the emergence of discrete boundaries among these representational regimes as functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). Varying feature sparsity and relative importance in the input distribution elicits qualitatively different behavior; for instance, greater feature sparsity can encourage increased superposition. By analyzing expert-specific phase diagrams in MoEs, we show that experts adopt representational strategies that differ from those of dense models.

Setup. We use the setup described in Section 4.1, adding a load-balancing loss in selected configurations to prevent specialization collapse, i.e. local minima in which the router and experts entirely ignore some experts. We consider three model configurations, each with one active expert ($k = 1$): (A) $n=2$, $m=1$; (B) $n=3$, $m=1$; and (C) $n=3$, $m=2$. For each configuration, we report expert-specific phase diagrams over feature sparsity and last-feature relative importance while varying network sparsity by increasing the number of experts (E) up to the number of input feature dimensions (n).

In this section, we fix active rather than total parameters, so that $m_{dense} = km_{moe}$ when router parameters are ignored. This choice enables comparisons within architectural families; otherwise, observed differences could be attributed to architectural variation rather than to the number of active parameters. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. The latter, however, has only one hidden dimension ($m = 1$) with which to encode the same number of input features ($n = 3$) as the former, which has two hidden dimensions ($m = 2$), making specialization harder to interpret purely through the lens of superposition.

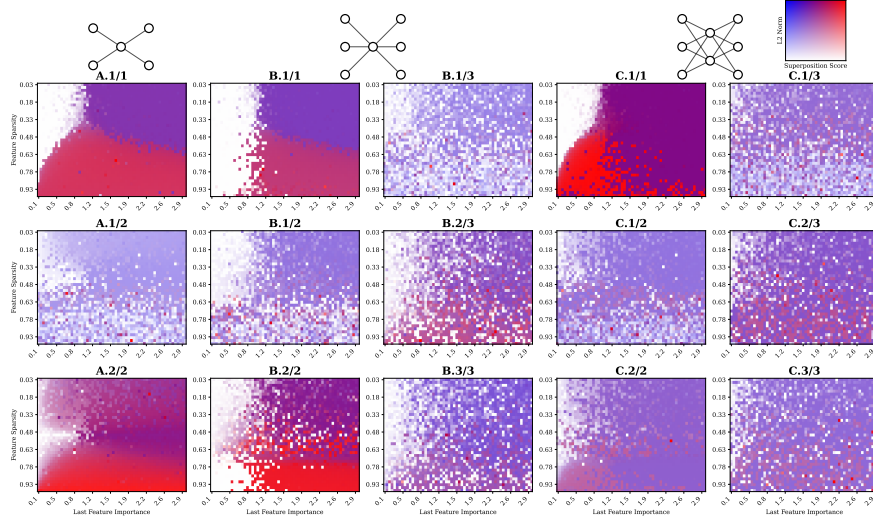


Figure 4: For a given expert and input dimension, or feature, we decode the feature’s embedding status in the hidden dimension: ignored (white), monosemantic (blue-purple), or superimposed (red). We plot joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j<n}(\hat{W}_n \cdot W_j)^2$) over feature sparsity $S \in [0, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last feature among n total features. For each cell, we train ten models and select the model with lowest loss; this section uses the load-balancing loss. We visualize joint feature norm and superposition for the last feature: low L2 norm ($\|W_n\|$) appears white, indicating that the model ignores the last feature, whereas low superposition appears blue-purple, indicating monosemantic representation. Red indicates representation in superposition. Cell (i, j) in subfigure X.e/E denotes expert e among E total experts trained with architecture X at last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes a dense model.

Results and Takeaways. Across all single-expert, effectively dense, cases, we observe a clear phase change (Figure 4.X.1/1), consistent with Elhage et al. (2022). As the total number of experts increases, however, these discrete phase boundaries disappear. Some $E = 2$ MoE experts resemble their corresponding dense cases (Figure 4.X.2/2), but their

transitions are visibly more continuous. In each configuration, one expert becomes more monosemantic and specializes in the feature with highest relative importance. Experts that diverge from the dense cases consistently exhibit much lower superposition scores, visually appearing bluer, which indicates more monosemantic representation. This result aligns with the preceding section: relative to dense counterparts, MoEs favor representational regimes with lower superposition.

In the $n = 2, m = 1$ configuration (Figure 4.A), the dense model fails to represent the last feature when feature sparsity is low, whereas the comparable MoE preserves that feature more strongly because the expert allocation provides sufficient capacity. With three input dimensions (Figure 4.B), this behavior does not appear in the MoEs, because experts superimpose the other two features and leave no remaining capacity for the third feature within a single hidden dimension. Unlike the other two configurations, architecture B lacks sufficient hidden-dimensional capacity, even with superposition in the high-sparsity regime, to represent all features. Nevertheless, we do not observe a clean phase change, except in the $0.5 - 0.7$ feature-sparsity region of 4.B.2/2, where the boundary is largely discrete but still mixed. For $m = 2$, the white region in the dense model (Figure 4.C.1/1), occurring in the mid- to low-feature-sparsity domain when the last feature is relatively less important than the others, corresponds to ignoring that feature. As network sparsity increases across the remaining panels of Figure 4.C, however, the models represent the last feature with increasing L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). Thus, dimensionality in the low-relative-importance region increases with network sparsity, as also demonstrated in Figure 3.

We also observe, in Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2, a feature-sparsity window from roughly 0.48 to 0.7 in which polysemanticity, monosemanticity, and feature ignorance are heavily intermixed. This suggests the existence of an intermediate MoE regime in which polysemantic and monosemantic solutions achieve comparable loss, making it difficult for training to commit consistently to the strategies observed in the low- and high-feature-sparsity domains. No analogous pattern appears in dense models, providing further evidence that MoEs learn distinct representational strategies.

Conclusion. All experiments in this section use $\text{top-}k = 1$ routing, so only one expert is active for any input. Even under this restriction, behavior changes substantially in the $E = 2$ case, including settings where hidden-dimensional capacity with superposition is sufficient to represent all features. We therefore conclude that it is misleading to regard MoEs as mere aggregations of dense models. The router, by restricting each expert to a subset of the feature domain, substantially alters both expert learning dynamics and the resulting representational geometry.

6 EXPERT SPECIALIZATION

Because MoEs exhibit reduced superposition, we next examine how these comparatively monosemantic features are organized within experts and how that organization relates to specialization.

Expert specialization in MoEs is often operationalized through load balancing across experts and inputs (Chaudhari et al., 2025). This criterion, however, does not fully capture the intuitive meaning of specialization, according to which an expert should be preferentially used when the concepts it specializes in are actually present in the input.

We therefore define an expert as specialized when it *occupies* particular feature directions in input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated and that, when they co-occur, they also encourage load balancing across experts.

Because we fix $k = 1$, the router partitions feature space into convex conic regions, with each region assigned to a particular expert (see Appendix A.3). By definition, $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained within the cone and s is any positive scalar. Hence, if a feature vector $\vec{x}$ is routed to an expert, every positive rescaling $s\vec{x}$ is

routed to the same expert. In this sense, we say that feature x is contained within expert e , and therefore that expert e *occupies* x .

Empirically, we find that small models distributing input space across more experts tend to achieve lower loss (see Appendix A.4). This observation motivates a further question: does allocation of input-space regions to particular experts predict representational properties of those experts? Our definition of specialization suggests that such allocation should be associated with *monosemanticity*, and we find that this correlation persists in larger toy models, for example when $m = 10$.

In models with $n > 2$, we explore whether initializing experts to occupy features in the input space cause the experts to be more monosemantic w.r.t. those features. Separately, we see if, for the features an expert has chosen to represent monosemantically, the expert occupies those features in the input space.

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied, and no additional feature, as shown in Figure 5a. Under ordered k-hot router initialization, the first expert monosemantically represents four of the five features it initially occupied, as shown in Figure 5b. The remaining experts, which were initialized over less important features, neither monosemantically represent those less important features nor recover the five most important features over which they were not initialized. When the ordering of feature importance is broken and each expert’s initially occupied features are randomized, each expert monosemantically represents only the most important feature or features among those it initially occupied, as shown in Figure 5c.

These results reveal a strong correlation between the features initially routed to an expert and the features that the same expert later represents monosemantically. We further observe that experts monosemantically represent only important features. This pattern holds both when each expert is explicitly initialized with a single important feature and when it is initialized with a feature set from which it effectively selects the most important member for monosemantic representation.

Under uniform feature importance, experts place all features into superposition and exhibit higher interference overall. Even in this regime, however, the features initially occupied by an expert remain *relatively* more monosemantic, achieving superposition scores that are, on average, 1.03 standard deviations below that expert’s mean.

We next test whether experts that monosemantically represent particular features also occupy those features in input space. To do so, we measure usage statistics when those features are *one of many* active features and when they are the *only* active features. The latter condition is equivalent to estimating the probability that the expert occupies the relevant feature direction. As Table 1 shows, the correlation holds under both xavier and k-hot initialization. Since $E = 10$, mean expert usage of $\sim 10\%$ indicates approximately even load balancing across experts. In all cases, expert usage increases substantially when the corresponding monosemantic feature or features are active, and when those features are the only active features, the associated expert dominates routing. Under k-hot initialization, *100%* of all features that an expert represents monosemantically are occupied by that same expert.

As experts represent more features monosemantically, they become more specialized in the feature-based sense defined above. Their usage on arbitrary inputs decreases, but conditional on the activation of their specialized features, their usage increases much more than that of other experts. This pattern holds in all cases except the xavier-initialized model containing an expert with four monosemantic features, where utilization drops substantially.

7 CONCLUSION

We investigated how experts modulate superposition in MoEs and found that, relative to dense networks, MoEs consistently exhibit greater monosemanticity while failing to display the same discontinuous phase changes. We also introduced a feature-based definition of

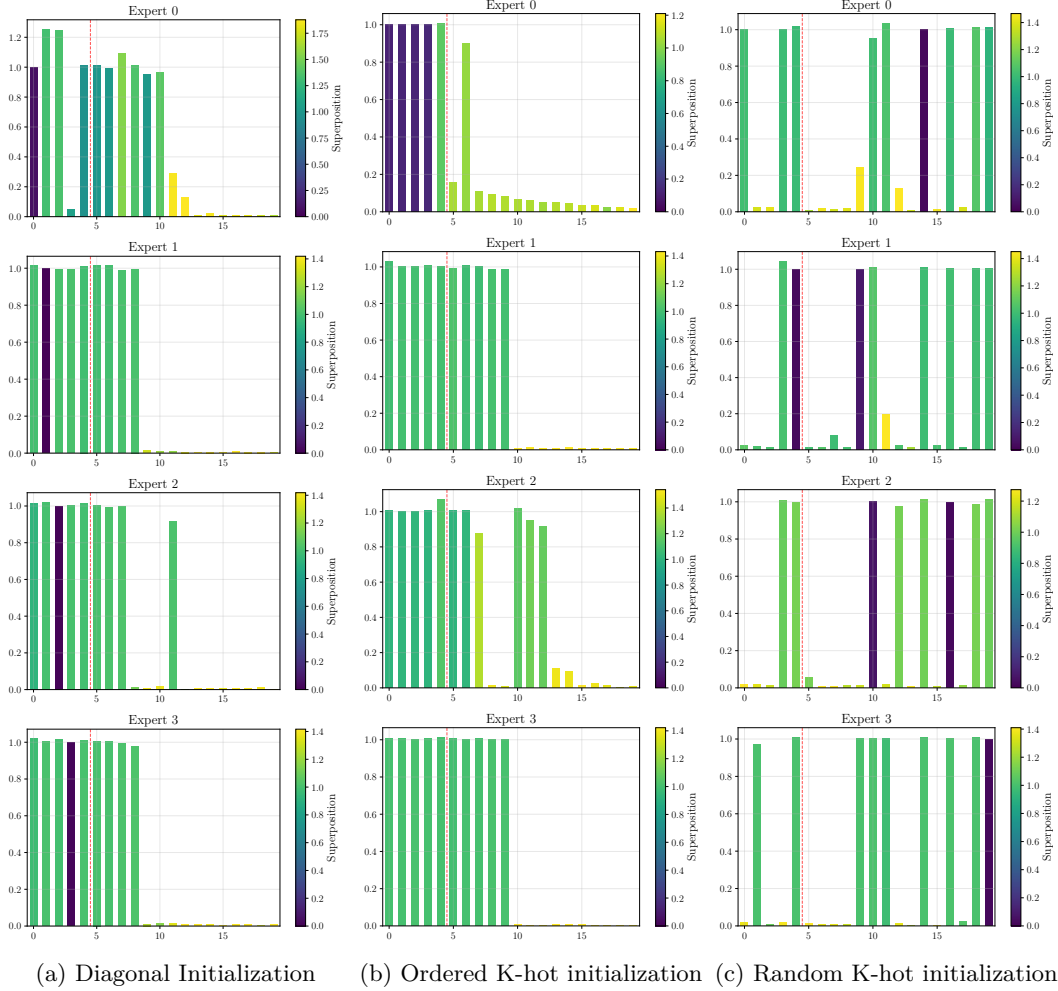


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition, indicated by color, under three initialization schemes with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), and relative feature importance decreases exponentially from feature one through feature 20. In (b), the gate matrix uses an “ordered k-hot” initialization, so that the first expert aligns with the first five features and each subsequent expert aligns with the next five features; relative feature importance is the same as in (a). In (c), the gate matrix uses a “random k-hot” initialization, assigning each expert five random features such that experts share no feature while collectively covering all 20 features. Relative feature importance again decreases exponentially, but its ordering is randomly distributed across features.

expert specialization and showed that, when initialization encourages specialization, experts naturally organize around coherent features. These findings are nevertheless derived from simple autoencoder toy models with synthetic data, leaving unresolved their generalization to large-scale transformers whose feature distributions are unknown (see Appendix A.6). Even with this caveat, our results show that toy MoEs can achieve comparable loss while maintaining more interpretable representations, thereby challenging the prevailing assumption that mechanistic interpretability and model capability must be fundamentally in tension. Future work should identify the conditions that favor monosemanticity in MoEs, characterize how MoE training dynamics diverge from those of dense models, and determine when specialization emerges. Progress on these questions may inform the design of language models that are both high-performing and more interpretable.

Table 1: Monosemantic feature and usage statistics per expert for $n = 100, m = 10, E = 10$. We train one hundred models for each initialization scheme, xavier and k-hot, yielding 1000 experts per scheme. Statistics are aggregated across models after classifying experts by the number of features they represent monosemantically. For the feature or features an expert represents monosemantically, we report expert usage when those features appear among several active input features, as well as expert usage when those features are the *only* active input features.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

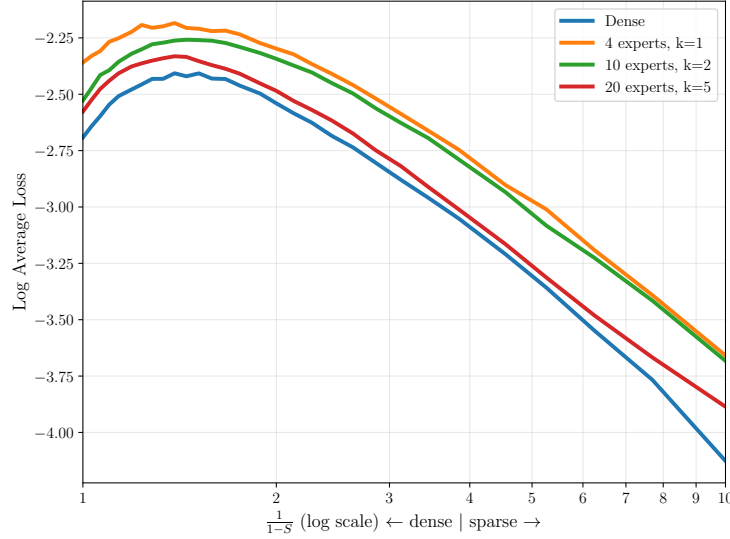


Figure 6: Log average loss as a function of feature density ($\frac{1}{1-s}$) for dense ($m = 20$), MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all using uniform feature importance ($I_i = 1.0$) over $n = 100$ input features. Results are averaged over five runs per sparsity level. Although the dense model outperforms all MoE models at every sparsity level, MoE loss approaches dense-model loss as expert count increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition arises as a way to exploit the mismatch between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, by contrast, structurally “eat this same gap” through conditional computation. By aligning activation sparsity with feature sparsity, MoEs reduce the computational penalty associated with dedicated, rarely activating neurons. As Elhage et al. (2022) note, when a model expends computation only on active features, decomposing polysemantic neurons into dedicated monosemantic ones becomes optimal, effectively exchanging the *compression* afforded by superposition for the *selection* afforded by routing.

Geometrically, this structural change appears as a reduction in *interference*. Although the global ratio of represented features to parameters may remain fixed, the router partitions feature space so that a feature routed to expert e competes only with the subset of features assigned to that same expert. Consequently, interference is governed by the local expert matrix $(W^e)^\top W^e$ rather than by the dense model’s global $W^\top W$. This partitioning sharply reduces the number of features that can interfere with any given feature vector, thereby weakening the optimization pressure toward superposition and enabling experts to learn more monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the $k = 1$ regime, the router function $g(x) = \text{softmax}(W^r x)$ is equivalent, for routing purposes, to $\text{argmax}(W^r x)$. The region routed to expert i can be written as $\forall j \neq i, (w_i - w_j)^\top x > 0$, where w_i and w_j are row vectors of W^r . This condition is a homogeneous linear inequality, and regions bounded by such inequalities are convex cones by definition. If a particular x satisfies the inequality, multiplying both sides by any positive scalar s preserves

satisfaction of the inequality. Likewise, if x_1 and x_2 both satisfy the inequality, then any $x = x_1\lambda + (1 - \lambda)x_2$ with $\lambda \in [0, 1]$ also satisfies it.

When $k > 1$, the set of inputs routed to a particular expert e becomes a union of convex cones rather than a single cone. Because such a union is not generally itself a convex cone, the interpretation of experts as occupying feature directions need not extend beyond the $k = 1$ setting.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

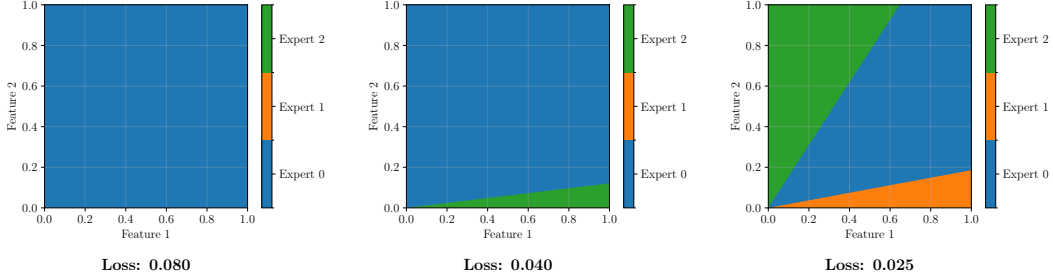


Figure 7: Expert routing for three otherwise identical models with different initialization schemes, using $n = 2$, $E = 3$, and $m = 1$. The first model (left) performs worst (loss: 0.08) and routes all inputs to a single expert. The second model (middle) performs better (loss: 0.04) and routes a small subset of inputs, specifically those for which feature 1 is active, to a second expert. The third model (right) achieves the lowest loss (loss: 0.025) and distributes input space across all experts: one expert is selected when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2, m = 1, E > 1$), tested over $E \in [2, 7]$, we empirically find that models distributing input space across more experts tend to achieve lower loss. Fixing $n = 2$ allows the routed regions of input space to be visualized directly, as shown in Figure 7.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setting of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical mapping from any dense model to a monosemantic MoE with an equivalent number of active features, provided that a sparsity constraint holds.

Assume that the number of active features is uniformly bounded by a , so that for any input $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Further assume that a does not exceed the hidden dimensionality m of an expert, thereby bounding the number of features any expert must represent, and that the hidden dimensionality is no larger than the total number of input features n ($a \leq m \leq n$). To construct the corresponding monosemantic MoE, create, for every possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$, an expert that monosemantically preserves the features in that subset. Equivalently, it suffices to take only subsets satisfying $|S| = a$. The router then selects the expert corresponding to the active-feature subset, which by assumption never contains more than a features:

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Because $|S| \leq a \leq m$, each expert has sufficient capacity to represent its assigned features without superposition. Thus, at most a features are active, and every unique active-feature combination is assigned to a dedicated expert with enough capacity to encode those features monosemantically. Under this construction, the number of possible experts required is $\binom{n}{a}$.

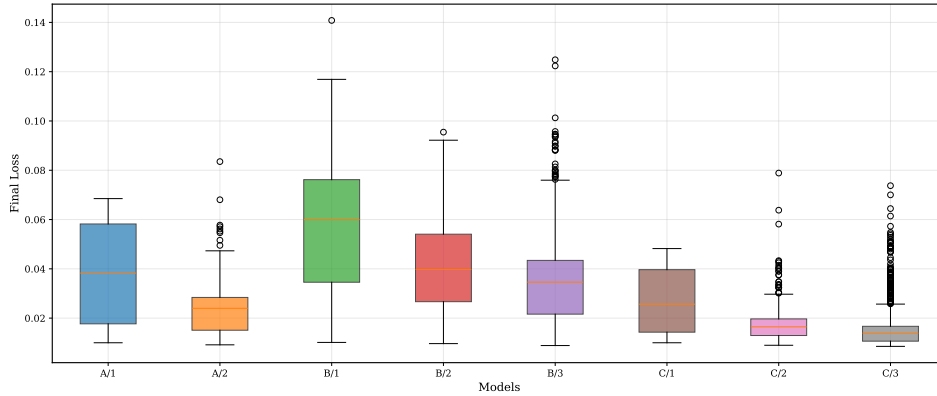


Figure 8: In model notation X/E , X denotes the architectures and models used in Figure 4, while E denotes the total number of experts, i.e. network sparsity. Increasing network sparsity lowers mean loss while increasing localized variance, particularly when the number of experts approaches the number of input feature dimensions. This pattern can be attributed to the comparatively unstable training of MoEs relative to dense models, despite training ten models for each cell and selecting the lowest-loss run.

This theoretical MoE achieves zero reconstruction loss only when the sparsity constraint holds, or approaches one in these toy models, because more than m features may otherwise be active simultaneously ($a \not\leq m$), exceeding the network’s monosemantic representational capacity. A dense polysemantic model would face the same limitation unless the feature distribution contained exploitable correlations. Consequently, even when $a \not\leq m$ occurs occasionally, the polysemantic model encounters an analogous bottleneck, and the monosemantic MoE constructed here may still outperform it under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ satisfying the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings studied here, this sparsity constraint holds in the limit as sparsity approaches one. In practice, however, there may be an upper bound on the number of features that a given quantity of information can semantically encode, as suggested by the size of meaningful embeddings for that data. Accordingly, a MoE with sufficiently many experts and a tractable, interpretable level of superposition may be adequate to encode all features present.

A.6 LIMITATIONS

Our results should be interpreted in light of several limitations. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). Although this design enables precise measurement of interference and monosemanticity, it abstracts away many complexities of large-scale transformers, including multiple computation, attention, heterogeneous feature distributions, and realistic routing dynamics. Consequently, the extent to which our findings transfer to real-world MoE architectures remains uncertain.

Second, our architectural and routing choices are intentionally restricted: we use simple experts, fixed top- k routing, often with $k = 1$, equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses except where explicitly noted. Practical MoEs often employ richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), and they frequently extend beyond reconstruction to objectives such as next-token prediction; these differences may induce qualitatively distinct representational behaviors.

Third, the feature distributions in our synthetic tasks, including their sparsity patterns, feature-importance profiles, and independence assumptions, are substantially simpler than the distributions encountered in natural data.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work suggests that these dynamics can change substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

Overall, this study provides mechanistic insight under clean experimental conditions, but further work is required to validate the same patterns in large, realistic MoE systems. Because our methods do not scale naturally to larger models, developing scalable analogues of these measurements is a clear direction for future work.